Algebra 1 — A Four-Week Course

Every lecture (short version) plus sample homework and exam problems with an answer key.

Week 1 — Expressions and linear equations

Real numbers; order of operations and expressions; properties and simplifying; one- and two-step equations; multi-step equations; literal equations and formulas; ratios, proportions, and percent.

1.1 Real numbers and operations

The number system algebra is built on.

Real numbers and operations

Algebra is arithmetic with letters standing in for numbers, so it begins with the numbers themselves. The **real numbers** include the **natural numbers** 1, 2, 3, ..., the **integers** (adding 0 and the negatives), the **rationals** (every fraction p/q with q ≠ 0), and the **irrationals** such as √2 and π, whose decimals never end or repeat.

Signed numbers

The **absolute value** |a| is the distance from a to 0 on the number line, so |−7| = |7| = 7. Adding a negative is subtracting, and subtracting a negative is adding. For products and quotients:

$$(-)(-) = (+),\ (-)(+) = (-).$$

Thus (−3)(−4) = 12 and −20 / 4 = −5. Division by zero is undefined: no number times 0 gives 5.

Opposites and reciprocals

Every number a has an **opposite** −a with a + (−a) = 0, and every nonzero number has a **reciprocal** 1/a with a · (1/a) = 1. Subtraction is adding the opposite; division is multiplying by the reciprocal. These two facts are the engine of equation solving in Lecture 1.4.

A real example

In 628 CE the Indian mathematician Brahmagupta wrote the Brahmasphutasiddhanta, the first text to treat zero as a number and to state rules for negatives. He called positive numbers fortunes and negative numbers debts: "a debt minus zero is a debt," "the product of two debts is a fortune." European mathematicians were still calling negative solutions "absurd" a thousand years later.

1.2 Order of operations and algebraic expressions

Reading an expression the only way it can be read.

Order of operations and algebraic expressions

An **algebraic expression** combines numbers, **variables**, and operations: $3x + 5$, $2(a - b)^2$. To **evaluate** one, substitute values for the variables and compute.

The order of operations

A single expression must have a single value, so a convention fixes the order:

- Grouping symbols first — parentheses, brackets, fraction bars, radicals;
- then exponents;
- then multiplication and division, left to right;
- then addition and subtraction, left to right.

$$3 + 4 \cdot 2^2 = 3 + 4 \cdot 4 = 19.$$

Translating words into symbols

"Five less than twice a number" is $2n - 5$, not $5 - 2n$. "The square of the sum of x and y" is $(x + y)^2$; "the sum of the squares" is $x^2 + y^2$. Most word-problem errors are translation errors, made before any algebra begins.

A real example

In 2019 the expression $8 \div 2(2 + 2)$ went viral, splitting millions of people between 16 and 1. Strict left-to-right gives 16; treating $2(2 + 2)$ as a single unit gives 1, and some older textbooks and calculators did exactly that. The dispute is a notation failure, not an arithmetic one — which is why programming languages require an explicit multiplication sign and why careful writers use parentheses.

1.3 Properties and simplifying expressions

The rules that let you rewrite an expression without changing it.

Properties and simplifying expressions

Two expressions are **equivalent** if they give the same value for every value of the variables. A few properties generate every legal rewrite.

The basic properties

- **Commutative**: $a + b = b + a$ and $ab = ba$;
- **Associative**: $(a + b) + c = a + (b + c)$ and $(ab)c = a(bc)$;
- **Identity**: $a + 0 = a$ and $a \cdot 1 = a$;
- **Distributive**: multiplication spreads over addition.

$$a(b + c) = ab + ac.$$

Combining like terms

Like terms have the same variables raised to the same powers: $4x$ and $-7x$ are like, $4x$ and $4x^2$ are not. The distributive property run backwards combines them: $4x - 7x = (4 - 7)x = -3x$. So $3(2x - 5) - (x - 4) = 6x - 15 - x + 4 = 5x - 11$. A minus sign in front of parentheses distributes -1 to every term inside.

A real example

Around 300 BCE Euclid's Elements, Book II, Proposition 1, proved the distributive law as geometry: a rectangle of height a cut into strips of widths b, c, and d has area equal to the sum of the strips' areas, $a(b + c + d) = ab + ac + ad$. Greek mathematicians had no algebraic notation, so every identity in this lecture was, for them, a statement about areas.

1.4 Solving one- and two-step equations

Undoing operations while keeping the balance.

Solving one- and two-step equations

An **equation** asserts that two expressions are equal. A **solution** is a value of the variable that makes the assertion true. Solving means producing simpler **equivalent equations** until the variable stands alone.

The properties of equality

Whatever you do to one side, do to the other: add, subtract, multiply, or divide (by a nonzero number) on both sides and the solutions are unchanged.

$$a = b \Rightarrow a + c = b + c \text{ and } ac = bc.$$

Two-step equations

Undo operations in reverse order: addition and subtraction first, then multiplication and division. For $3x - 7 = 11$, add 7 to get $3x = 18$, then divide by 3 to get $x = 6$. Check by substituting: $3(6) - 7 = 11$. Checking catches arithmetic errors at a cost of seconds.

A real example

The Rhind Mathematical Papyrus, copied by the Egyptian scribe Ahmes around 1550 BCE, contains "aha" problems — the first recorded linear equations. Problem 24 reads: a quantity and its seventh, added together, become 19. In modern symbols $x + x/7 = 19$, so $8x/7 = 19$ and $x = 16\ 5/8$. Ahmes reached the same answer by guessing 7, seeing it gave 8, and scaling up — the method of false position.

1.5 Multi-step equations

Variables on both sides, grouping, and fractions.

Multi-step equations

Harder equations need several moves, but the plan never changes: simplify each side, collect the variable on one side, then isolate it.

The procedure

- Clear fractions by multiplying both sides by the least common denominator.
- Distribute and combine like terms on each side.
- Add or subtract to move variable terms to one side and constants to the other.
- Divide by the coefficient, then check.

$$5x + 3 = 2x - 9 \Rightarrow 3x = -12 \Rightarrow x = -4.$$

No solution and infinitely many

Sometimes the variable cancels. If a false statement remains (2x + 1 = 2x + 5 gives 1 = 5), there is **no solution**. If a true statement remains (2(x + 3) = 2x + 6 gives 6 = 6), the equation is an **identity** and every real number is a solution.

A real example

A puzzle from the Greek Anthology, around 500 CE, gives the life of the mathematician Diophantus: boyhood lasted 1/6 of his life, youth 1/12 more, then 1/7 more before marriage; a son was born 5 years later and lived half as long as his father, who died 4 years after the son. So x = x/6 + x/12 + x/7 + 5 + x/2 + 4. Multiplying by 84 clears the fractions: 84x = 75x + 756, so x = 84.

1.6 Literal equations and formulas

Solving for one letter in terms of the others.

Literal equations and formulas

A **literal equation** has several variables. Solving it for one of them uses exactly the moves of Lecture 1.5, treating the other letters as if they were numbers.

Rearranging formulas

Distance equals rate times time, d = rt, solved for t gives t = d/r. The area of a trapezoid A = ½h(b_1 + b_2) solved for h gives h = 2A/(b_1 + b_2). Temperature:

$$F = (9/5)C + 32 \Rightarrow C = (5/9)(F - 32).$$

Units as a check

Units obey algebra too. If d is in miles and r in miles per hour, then d/r has units miles ÷ (miles/hour) = hours. A rearranged formula whose units come out wrong has been rearranged wrongly.

A real example

In 1983 Air Canada Flight 143, a new Boeing 767, ran out of fuel at 41,000 feet over Manitoba. The crew needed 22,300 kilograms of fuel and computed the volume to load from volume = mass ÷ density, but used 1.77 — the density in pounds per liter, not kilograms per liter. The plane took off with less than half the fuel required. The captain glided it to a safe landing at a former air base in Gimli.

1.7 Ratios, proportions, and percent

Comparing quantities by division.

Ratios, proportions, and percent

A **ratio** compares two quantities by division, a : b or a/b. A **rate** is a ratio of quantities in different units, such as 60 miles per hour. A **proportion** states that two ratios are equal.

Solving proportions

Multiplying both sides of a/b = c/d by bd gives the **cross-product** rule:

$$a/b = c/d \Rightarrow ad = bc.$$

If 3 pounds cost $7.50, then 3/7.50 = 8/x gives 3x = 60 and x = $20 for 8 pounds.

Percent

Percent means per hundred: 35% = 0.35. The percent equation is part = percent × whole. Percent change is (new − old)/old. A 20% increase followed by a 20% decrease is not a return to start: 1.2 × 0.8 = 0.96, a net loss of 4%.

A real example

Around 240 BCE Eratosthenes learned that at noon on the summer solstice the sun shone straight down a well at Syene, while at Alexandria, about 5,000 stadia north, it cast a shadow at an angle of about 7.2°. Since 7.2° is 1/50 of a full circle, the proportion 7.2/360 = 5,000/C gives Earth's circumference as 250,000 stadia — close to the true value, from one shadow and one ratio.

Week 2 — Inequalities, functions, and linear graphs

Linear inequalities; compound and absolute-value statements; relations and functions; slope; slope-intercept and point-slope forms; standard form and parallel and perpendicular lines; linear models.

2.1 Linear inequalities

Solving when the answer is a range, not a number.

Linear inequalities

An **inequality** compares expressions with <, >, ≤, or ≥. Its solution is usually an infinite set of numbers, drawn as a ray on the number line — an open circle for < or >, a closed circle for ≤ or ≥.

The one new rule

Inequalities are solved like equations, with one exception: multiplying or dividing both sides by a **negative** number reverses the direction of the inequality.

$$-3x > 12 \Rightarrow x < -4.$$

The reason: $-3 < 2$, but multiplying by -1 gives $3 > -2$. Negation reflects the number line and reverses order.

Interval notation

The set $x < -4$ is written $(-\infty, -4)$; the set $x \geq 2$ is $[2, \infty)$. Parentheses exclude an endpoint and brackets include it. Word problems use "at least" for ≥ and "at most" for ≤.

A real example

The symbols < and > first appeared in print in 1631, in Artis Analyticae Praxis, a book by the English mathematician Thomas Harriot published ten years after his death. Harriot, who had sailed to Roanoke as Walter Raleigh's scientific adviser, also drew the moon through a telescope months before Galileo. The ≤ and ≥ signs came a century later, from the French mathematician Pierre Bouguer.

2.2 Compound and absolute-value statements

Two conditions at once, and distance from a target.

Compound and absolute-value statements

A **compound inequality** joins two inequalities. "And" requires both, giving an overlap such as $-2 < x \leq 5$. "Or" requires at least one, giving two pieces such as $x < -1$ or $x > 3$.

Absolute-value equations

Since |x| is distance from 0, |x − c| is distance from c. An equation |A| = k with k > 0 splits into two cases, A = k or A = −k. So |2x − 1| = 7 gives x = 4 or x = −3. If k < 0 there is no solution.

Absolute-value inequalities

Distance less than k is an "and"; distance greater than k is an "or":

$$|A| < k \Leftrightarrow -k < A < k,\ |A| > k \Leftrightarrow A < -k \text{ or } A > k.$$

Thus |x − 10| ≤ 3 means 7 ≤ x ≤ 13: every number within 3 of 10.

A real example

Manufacturing runs on absolute-value inequalities. A shaft specified as 25.00 ± 0.05 millimeters must satisfy |d − 25.00| ≤ 0.05. Such tolerances make interchangeable parts possible — an idea the French gunsmith Honoré Blanc demonstrated in 1785 by assembling musket locks from randomly chosen parts, a show that so impressed Thomas Jefferson that he urged it on the United States.

2.3 Relations and functions

Rules that assign each input exactly one output.

Relations and functions

A **relation** is any set of input–output pairs. A **function** is a relation in which each input has exactly one output. The set of inputs is the **domain**; the set of outputs is the **range**.

Function notation

Writing f(x) names the function f and its input x. If $f(x) = 2x^2 - 3$, then

$$f(4) = 2(16) - 3 = 29.$$

The notation f(x) does not mean f times x. It means "the output of f when the input is x."

Representations and the vertical line test

A function can be given by a table, an equation, a graph, or words. A graph represents a function exactly when every vertical line crosses it at most once; the circle $x^2 + y^2 = 25$ fails, since x = 3 meets it at y = 4 and y = −4. The **independent variable** is the input; the **dependent variable** is the output.

A real example

Gottfried Leibniz introduced the word "function" in the 1690s, and in 1734 Leonhard Euler introduced the notation f(x). Euler's notation spread through his textbooks, which taught Europe mathematics for a century. The modern definition — any rule assigning one output to each input, formula or not — was settled by Peter Gustav Lejeune Dirichlet in 1837.

2.4 Slope and rate of change

How steep a line is, as a number.

Slope and rate of change

The **slope** of a line measures its steepness: the vertical change per unit of horizontal change. Between two points (x_1, y_1) and (x_2, y_2):

The slope formula

$$m = (y_2 - y_1) / (x_2 - x_1).$$

Rise over run. From (1, 2) to (4, 11) the slope is 9/3 = 3: y rises 3 for every 1 that x increases.

Reading slope

Positive slope rises left to right; negative slope falls; zero slope is horizontal; a vertical line has **undefined** slope, since its run is zero. In applications slope is a **rate of change** with units — dollars per hour, miles per gallon, degrees per minute. A function has constant rate of change exactly when its graph is a line.

A real example

Accessibility standards issued under the Americans with Disabilities Act of 1990 cap wheelchair ramps at a slope of 1/12 — one inch of rise for every twelve inches of run. A ramp climbing to a 30-inch porch must therefore run at least 30 × 12 = 360 inches, or 30 feet. Road signs state the same idea as a percent: a 6% grade rises 6 feet per 100 feet of road.

2.5 Slope-intercept and point-slope forms

Writing the equation of a line.

Slope-intercept and point-slope forms

Every nonvertical line is the graph of a linear function. Two forms of its equation cover nearly every use.

Slope-intercept form

$$y = mx + b.$$

Here m is the slope and b is the **y-intercept**, where the line crosses the y-axis. To graph $y = -\frac{2}{3}x + 4$, plot (0, 4), then move down 2 and right 3. In a model, b is the starting value and m the rate: a taxi charging $3 plus $2.50 per mile costs $y = 2.5x + 3$.

Point-slope form

Given a slope m and any point (x_1, y_1) on the line, $y - y_1 = m(x - x_1)$. Given two points, find m with the slope formula, then use either point. This is the fastest route from data to an equation.

A real example

Water freezes at 0°C and 32°F and boils at 100°C and 212°F. These two points fix a line: its slope is (212 − 32)/(100 − 0) = 1.8, and point-slope form through (0, 32) gives F = 1.8C + 32. Every thermometer conversion is this line. Setting F = C shows the two scales agree at exactly −40°.

2.6 Standard form, parallel and perpendicular lines

Other forms, and how lines relate.

Standard form, parallel and perpendicular lines

A line can also be written in **standard form** Ax + By = C. It is the natural form for intercepts: set y = 0 to find the x-intercept and x = 0 to find the y-intercept. For 3x + 4y = 12 these are (4, 0) and (0, 3).

Special lines

A **horizontal line** has equation y = k and slope 0. A **vertical line** has equation x = h and undefined slope; it is not a function.

Parallel and perpendicular

Distinct nonvertical lines are **parallel** exactly when their slopes are equal. They are **perpendicular** exactly when their slopes are negative reciprocals:

$$m_1 = m_2 \text{ (parallel)}, \; m_1 \cdot m_2 = -1 \text{ (perpendicular)}.$$

The perpendicular to y = 2x + 1 through (4, 0) has slope −½, so y = −½(x − 4).

A real example

In 1637 René Descartes published La Géométrie as an appendix to his Discourse on Method. It joined algebra to geometry by labeling points with number pairs, so that curves become equations and geometric questions become algebra problems. Whether two lines are perpendicular — once a matter of compass and straightedge — became a one-line check on their slopes.

2.7 Linear models and scatter plots

Fitting a line to real, messy data.

Linear models and scatter plots

Real data never fall exactly on a line. A **scatter plot** shows paired data as points; if they cluster around a line, a **linear model** summarizes the trend.

Correlation and the line of best fit

The **correlation coefficient** r, between −1 and 1, measures how closely the points hug a line: near ±1 is strong, near 0 is weak. The **line of best fit**, or least-squares line, minimizes the sum of the squared vertical distances from the points to the line:

$$\hat{y} = mx + b, \text{ minimizing } \Sigma\,(y - \hat{y})^2.$$

Using and misusing a model

Predicting inside the range of the data is **interpolation**; predicting beyond it is **extrapolation**, which is riskier. The slope interprets as a rate and the intercept as a starting value — when either makes sense. And correlation is not causation: ice cream sales and drownings rise together because both follow summer.

A real example

In 1929 Edwin Hubble plotted the speeds of 24 galaxies against their distances. The points scattered widely, but a line through them showed speed proportional to distance, $v = H_0 d$. The slope H_0 — the Hubble constant — measures how fast the universe is expanding, and its reciprocal estimates the age of the universe. Hubble's slope was about 500; the modern value is about 70.

Week 3 — Systems, exponents, and polynomials

Systems by graphing; substitution and elimination; systems of inequalities; laws of exponents; scientific notation and exponential functions; polynomial operations; special products.

3.1 Systems of equations by graphing

Where two lines meet.

Systems of equations by graphing

A **system of linear equations** is two or more equations considered together. A solution satisfies every equation at once; for two lines, it is their point of intersection.

Three possibilities

- Lines with different slopes meet once: **one solution**.
- Parallel lines never meet: **no solution** (inconsistent).
- Identical lines meet everywhere: **infinitely many solutions** (dependent).

$$y = 2x - 1,\ y = -x + 5 \Rightarrow (2,\ 3).$$

Strengths and limits of graphing

Graphing shows at a glance which case you are in, but reading a point such as (1.37, 2.84) off a sketch is imprecise. Use graphs to see and algebra (Lecture 3.2) to compute. Break-even problems are the classic application: cost and revenue lines cross where profit is zero.

A real example

Before satellite navigation, a ship's navigator fixed position by taking compass bearings on two known landmarks and drawing each as a line on the chart. The ship lay on both lines, so it lay at their intersection — a "fix." A third bearing served as a check: if the three lines formed a large triangle rather than meeting at a point, a bearing was wrong.

3.2 Substitution and elimination

Solving systems exactly.

Substitution and elimination

Two algebraic methods solve any system exactly by reducing two equations in two unknowns to one equation in one unknown.

Substitution

Solve one equation for one variable and substitute into the other. From $y = 3x$ and $x + y = 12$: $x + 3x = 12$, so $x = 3$ and $y = 9$. Best when a variable already has coefficient 1.

Elimination

Add or subtract the equations, after multiplying one or both by constants, so that one variable cancels:

$$2x + 3y = 12,\ 4x - 3y = 6 \Rightarrow 6x = 18 \Rightarrow x = 3,\ y = 2.$$

If both variables cancel, a false result such as $0 = 7$ means no solution; $0 = 0$ means infinitely many. Mixture, investment, and speed problems all reduce to systems solved this way.

A real example

The Chinese text Nine Chapters on the Mathematical Art, compiled around the first century CE, devotes its eighth chapter, Fangcheng, to systems of linear equations. Coefficients were laid out as columns of counting rods on a board, and one column was repeatedly multiplied and subtracted from another until unknowns were eliminated — the method Europe would credit to Carl Friedrich Gauss some 1,700 years later.

3.3 Systems of linear inequalities

Regions of the plane, and the best point in them.

Systems of linear inequalities

The graph of a linear inequality such as $y > 2x - 3$ is a **half-plane**: the boundary line, dashed for $<$ or $>$ and solid for $\le$ or $\ge$, plus the shaded side that satisfies the inequality.

Graphing a system

Test a point not on the boundary — $(0, 0)$ when possible — to choose the side to shade. The solution of a system is the overlap of all the half-planes, the **feasible region**.

Linear programming

To maximize or minimize a linear quantity such as profit $P = ax + by$ over a feasible region bounded by lines, check only the corners:

$$\text{the optimum of } P = ax + by \text{ occurs at a vertex.}$$

A bakery limited by oven hours and flour finds its most profitable mix of bread and cake by evaluating profit at each corner of the region its constraints enclose.

A real example

In 1945 the economist George Stigler asked for the cheapest diet meeting nine nutritional requirements from 77 foods — a system of linear inequalities. With no general method he found a near-optimal answer of $39.93 a year at 1939 prices. In 1947 George Dantzig invented the simplex method, which walks from corner to corner of the feasible region; applied to Stigler's problem, it found the true minimum, $39.69.

3.4 Laws of exponents

Shorthand for repeated multiplication, and its rules.

Laws of exponents

For a positive integer n, a^n means a multiplied by itself n times; a is the **base** and n the **exponent**. Counting factors proves the laws.

The laws

$$a^m a^n = a^{m+n},\ a^m/a^n = a^{m-n},\ (a^m)^n = a^{mn}.$$

Also $(ab)^n = a^n b^n$ and $(a/b)^n = a^n/b^n$. But $(a + b)^n$ is **not** $a^n + b^n$: exponents distribute over products, not sums.

Zero and negative exponents

Keeping the quotient law true forces the definitions $a^0 = 1$ and $a^{-n} = 1/a^n$ for $a \neq 0$. So $2^{-3} = 1/8$ and $5^0 = 1$. A negative exponent means reciprocal, never a negative number.

A real example

Around 250 BCE Archimedes wrote The Sand Reckoner to show that the number of grains of sand needed to fill the universe, though enormous, is finite. Greek numerals stopped at a myriad, 10,000, so he built a system of orders of magnitude and proved that multiplying numbers adds their orders — in modern terms $a^m a^n = a^{m+n}$. His estimate came to about 10^{63} grains.

3.5 Scientific notation and exponential functions

Very large, very small, and very fast.

Scientific notation and exponential functions

Scientific notation writes a number as $a \times 10^n$ with $1 \le a < 10$: the speed of light is 3.0×10^8 m/s, a hydrogen atom about 1×10^{-10} m wide. The laws of exponents multiply such numbers: $(2 \times 10^3)(4 \times 10^5) = 8 \times 10^8$.

Exponential functions

In a linear function a fixed amount is **added** each step; in an **exponential function** a fixed factor **multiplies**:

$$y = a \cdot b^x,\ b > 0,\ b \neq 1.$$

Here a is the initial value. If $b > 1$ the function shows **growth**; if $0 < b < 1$, **decay**.

Growth and decay rates

A quantity growing r percent per period is $y = a(1 + r)^t$; decaying, $y = a(1 - r)^t$. \$1,000 at 5% compound interest grows to $1{,}000(1.05)^{10} \approx \$1{,}629$ in ten years. Exponential growth always eventually overtakes linear growth, however large the linear slope.

A real example

In 1965 Gordon Moore, later a co-founder of Intel, observed that the number of transistors on a chip had doubled every year, and predicted it would continue; in 1975 he revised the period to two years. Doubling every two years is $y = a \cdot 2^{t/2}$. Moore's law held for about five decades, carrying chips from dozens of transistors to tens of billions.

3.6 Polynomials and their operations

Adding, subtracting, and multiplying expressions.

Polynomials and their operations

A **polynomial** is a sum of terms of the form ax^n with whole-number exponents, such as $4x^3 - 2x + 7$. Its **degree** is the highest exponent; the term with that exponent is the **leading term**. Polynomials are named by degree (linear, quadratic, cubic) and by number of terms (monomial, binomial, trinomial).

Adding and subtracting

Combine like terms. To subtract, distribute the minus sign to every term of the second polynomial: $(5x^2 + 3x - 1) - (2x^2 - x + 4) = 3x^2 + 4x - 5$.

Multiplying

Multiply every term of the first by every term of the second — the distributive property repeated. For two binomials this is FOIL (first, outer, inner, last):

$$(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = x^2 - 2x - 15.$$

The degree of a product is the sum of the degrees.

A real example

In 1591 the French lawyer François Viète published In Artem Analyticem Isagoge, the first work to use letters systematically for both unknowns (vowels) and known quantities (consonants). Before Viète every problem was stated with specific numbers; afterward a single expression could describe infinitely many problems at once, and a polynomial became an object that could itself be manipulated.

3.7 Special products

Patterns worth memorizing.

Special products

Three products occur so often that recognizing them saves time in both directions — multiplying now, factoring in Week 4.

Squares of binomials

$$(a + b)^2 = a^2 + 2ab + b^2,\ (a - b)^2 = a^2 - 2ab + b^2.$$

The middle term 2ab is exactly what is forgotten when students write $(a + b)^2 = a^2 + b^2$. Mental arithmetic: $51^2 = 2{,}500 + 100 + 1 = 2{,}601$.

Sum times difference

$(a + b)(a - b) = a^2 - b^2$: the middle terms cancel. So $49 \times 51 = 50^2 - 1 = 2{,}499$. Higher powers of a + b follow Pascal's triangle, each row formed by adding adjacent entries of the row above: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$.

A real example

The triangle of binomial coefficients is named for Blaise Pascal, whose treatise on it was written in 1654. But the Chinese mathematician Yang Hui printed it in 1261, crediting Jia Xian, who used it around 1050; in China it is still called Yang Hui's triangle. The Persian poet and mathematician Omar Khayyam also knew the binomial expansions in the eleventh century.

Week 4 — Factoring, quadratics, and radicals

Factoring by GCF and grouping; factoring trinomials; solving quadratics by factoring; graphing quadratic functions; completing the square and the quadratic formula; radicals and the Pythagorean theorem; rational expressions; capstone synthesis.

4.1 Factoring: GCF and grouping

Running multiplication in reverse.

Factoring: GCF and grouping

To **factor** a polynomial is to write it as a product. Factoring undoes multiplication and is the key to solving polynomial equations.

The greatest common factor

Always begin by factoring out the **greatest common factor** of all terms — the largest number and highest power of each variable dividing every term:

$$6x^3 + 9x^2 - 15x = 3x(2x^2 + 3x - 5).$$

Factoring by grouping

With four terms, group them in pairs, factor each pair, and look for a common binomial factor: $x^3 + 2x^2 + 3x + 6 = x^2(x + 2) + 3(x + 2) = (x^2 + 3)(x + 2)$. Check any factorization by multiplying back.

A real example

Around 300 BCE Euclid's Elements, Book VII, gave an algorithm for the greatest common divisor of two numbers: repeatedly replace the larger by the remainder when it is divided by the smaller, until the remainder is zero. For 252 and 105: $252 = 2 \cdot 105 + 42$, $105 = 2 \cdot 42 + 21$, $42 = 2 \cdot 21$, so the GCD is 21. The Euclidean algorithm is still used billions of times a day inside encryption software.

4.2 Factoring trinomials

Finding the two binomials that multiply to a quadratic.

Factoring trinomials

A quadratic trinomial $ax^2 + bx + c$ often factors into two binomials. Factoring means reversing FOIL.

When a = 1

Find two numbers whose product is c and whose sum is b:

$$x^2 + bx + c = (x + p)(x + q),\ pq = c,\ p + q = b.$$

For $x^2 - x - 12$ you need product -12 and sum -1: the pair -4 and 3, so $(x - 4)(x + 3)$. A positive c means the two numbers share b's sign; a negative c means they have opposite signs.

When a ≠ 1, and special patterns

Find two numbers with product ac and sum b, split the middle term, and factor by grouping: $6x^2 + 11x + 4$ has ac = 24, split as 3 and 8, giving $(3x + 4)(2x + 1)$. Recognize the special products of Lecture 3.7 in reverse: $a^2 - b^2 = (a + b)(a - b)$, and $a^2 \pm 2ab + b^2 = (a \pm b)^2$. A polynomial with no factorization over the integers is **prime**.

A real example

Factoring is easy to check and hard to do: multiplying two large primes takes a computer a moment, but recovering them from their product can take longer than the age of the universe. In 1977 Ron Rivest, Adi Shamir, and Leonard Adleman built the RSA encryption system on exactly this asymmetry. Every secure web connection rests on the difficulty of factoring.

4.3 Solving quadratics by factoring

The zero-product property.

Solving quadratics by factoring

A **quadratic equation** has the form $ax^2 + bx + c = 0$ with $a \neq 0$. Its solutions are also called **roots** or **zeros**.

The zero-product property

If a product is zero, at least one factor is zero:

$$ab = 0 \Rightarrow a = 0 \text{ or } b = 0.$$

So to solve a quadratic, move everything to one side, factor, and set each factor equal to zero. From $x^2 - x - 12 = 0$: $(x - 4)(x + 3) = 0$, so $x = 4$ or $x = -3$.

The classic error

The property works only for zero. From $(x - 2)(x - 3) = 6$ you may not conclude $x - 2 = 6$: expand to $x^2 - 5x = 0$, then factor as $x(x - 5) = 0$, so $x = 0$ or $x = 5$. Dividing both sides by x loses the solution $x = 0$ — factor instead.

A real example

A Babylonian clay tablet from about 1800 BCE, now in the British Museum as BM 13901, poses the problem: "I added the area and the side of my square: 3/4." In modern terms $x^2 + x = 3/4$, which factors as $(x - 1/2)(x + 3/2) = 0$. The scribe's recipe produces the answer 1/2. Negative lengths made no sense, so the second root, $-3/2$, was never considered.

4.4 Graphing quadratic functions

Parabolas: vertex, axis, and intercepts.

Graphing quadratic functions

The graph of $f(x) = ax^2 + bx + c$ is a **parabola**. It opens upward if $a > 0$ and downward if $a < 0$; a larger $|a|$ makes it narrower.

Vertex and axis of symmetry

The parabola is symmetric about a vertical line through its **vertex**, the lowest or highest point:

$$x = -b / (2a).$$

For $f(x) = x^2 - 4x + 1$ the axis is $x = 2$ and the vertex is $(2, f(2)) = (2, -3)$, the minimum.

Vertex form and intercepts

Vertex form $f(x) = a(x - h)^2 + k$ shows the vertex (h, k) directly, as a shift of $y = x^2$. The y-intercept is c; the x-intercepts are the roots of the equation $f(x) = 0$. A parabola may cross the x-axis twice, touch it once at the vertex, or miss it entirely — matching two, one, or no real solutions.

A real example

In 1638 Galileo Galilei published Two New Sciences, proving that a projectile's path is a parabola: horizontal motion at constant speed combined with vertical fall that grows with the square of time. A ball thrown upward at 48 feet per second from 6 feet has height $h = -16t^2 + 48t + 6$. The vertex, at $t = 48/32 = 1.5$ seconds, gives the peak height, 42 feet.

4.5 Completing the square and the quadratic formula

Solving every quadratic.

Completing the square and the quadratic formula

Many quadratics do not factor over the integers. Two methods solve them all.

Completing the square

Since $(x + k)^2 = x^2 + 2kx + k^2$, add the square of half the x-coefficient to both sides. For $x^2 + 6x = 7$: add 9 to get $(x + 3)^2 = 16$, so $x + 3 = \pm 4$ and $x = 1$ or $x = -7$. The same move converts standard form to vertex form.

The quadratic formula

Completing the square on $ax^2 + bx + c = 0$ in general gives

$$x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a).$$

The **discriminant** $b^2 - 4ac$ decides the number of real solutions: positive, two; zero, one; negative, none, since no real number has a negative square.

A real example

Around 820 CE Muhammad ibn Musa al-Khwarizmi, working at the House of Wisdom in Baghdad, solved $x^2 + 10x = 39$ by literally completing a square: a square of side x with two rectangles 5 by x attached, then a missing corner of area 25 filled in to make a larger square of area 64 and side 8, so x = 3. The title of his book, al-jabr, gave us the word algebra, and his name gave us algorithm.

4.6 Radicals and the Pythagorean theorem

Square roots and right triangles.

Radicals and the Pythagorean theorem

The **principal square root** $\sqrt{a}$ of $a \geq 0$ is the nonnegative number whose square is a. Numbers such as 49 are **perfect squares**; the square root of any other positive integer is irrational.

Simplifying radicals

$$\sqrt{(ab)} = \sqrt{a} \cdot \sqrt{b},\ \sqrt{(a/b)} = \sqrt{a} / \sqrt{b}.$$

Factor out perfect squares: $\sqrt{72} = \sqrt{36} \cdot \sqrt{2} = 6\sqrt{2}$. Add only like radicals, $3\sqrt{5} + 4\sqrt{5} = 7\sqrt{5}$. **Rationalize** a denominator by multiplying top and bottom by the radical: $1/\sqrt{2} = \sqrt{2}/2$. Note $\sqrt{(a + b)} \neq \sqrt{a} + \sqrt{b}$.

The Pythagorean theorem

In a right triangle with legs a and b and hypotenuse c, $a^2 + b^2 = c^2$. It gives the **distance formula** $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$. A 3-4-5 triangle is right; carpenters check a square corner by measuring 3 feet, 4 feet, and a 5-foot diagonal.

A real example

A Babylonian tablet from about 1800 BCE, YBC 7289 at Yale, shows a square with its diagonals and the number 1;24,51,10 in base 60 — about 1.41421296. The true value of $\sqrt{2}$ is 1.41421356..., so the scribe was correct to six decimal places. The tablet shows that the diagonal of a unit square is $\sqrt{2}$, a Pythagorean fact known more than a thousand years before Pythagoras.

4.7 Rational expressions

Algebraic fractions.

Rational expressions

A **rational expression** is a quotient of polynomials, such as $(x + 1)/(x^2 - 4)$. It is undefined wherever its denominator is zero — here at x = 2 and x = −2 — and those values are excluded from its domain.

Simplifying and operating

Factor numerator and denominator, then cancel common **factors** — never common terms:

$$(x^2 - 9)/(x^2 + 3x) = (x + 3)(x - 3) / [x(x + 3)] = (x - 3)/x.$$

Multiply and divide as with numerical fractions. To add or subtract, rewrite over a common denominator.

Rational equations and inverse variation

Multiply both sides by the least common denominator to clear fractions, solve, and then discard any **extraneous** solution that makes an original denominator zero. When y = k/x, y **varies inversely** with x: doubling one halves the other. Work problems combine rates: if one pipe fills a tank in 3 hours and another in 6, together they fill 1/3 + 1/6 = 1/2 tank per hour, so 2 hours.

A real example

Every camera focuses by a rational equation. A lens of focal length f images an object at distance d_o at distance d_i behind it, where $1/f = 1/d_o + 1/d_i$. For a 50 mm lens and a subject 2,000 mm away, $1/d_i = 1/50 - 1/2{,}000 = 39/2{,}000$, so $d_i \approx 51.3$ mm. Autofocus moves the lens those 1.3 mm.

4.8 Capstone synthesis

One idea behind everything: equals done to equals stay equal.

Capstone synthesis

Step back, and the entire course reduces to a single sentence: algebra replaces numbers with letters and solves for the unknown by transforming equations without changing their solutions.

The one principle

$$a = b \Rightarrow f(a) = f(b).$$

Everything you have learned elaborates this statement:

- Expressions and equations are rewritten and solved by equivalent steps (Week 1).
- Functions and their graphs turn equations into pictures, and lines into rates (Week 2).
- Systems, exponents, and polynomials extend the same moves to harder objects (Week 3).
- Factoring and the quadratic formula solve every quadratic equation (Week 4).

Where algebra goes next

Algebra 2 extends these ideas to complex numbers, higher-degree polynomials, logarithms, and conic sections. Geometry, trigonometry, statistics, and calculus all rest on everything here — calculus, in particular, is algebra applied to limits.

A real example

The quadratic formula has cousins for degree 3 and 4, published by Gerolamo Cardano in 1545. For degree 5 mathematicians searched for three centuries, until Niels Henrik Abel proved in 1824 that no such formula in radicals exists. Évariste Galois, who died in a duel at twenty, explained exactly which equations can be solved — founding abstract algebra, the subject that grew from the question of when x can be isolated.

Practice problems

A sample of the graded work in the course: two homeworks plus the midterm and final. Answers follow at the end.

Homework 1.1 — Expressions and equations

Short-answer problems on expressions and linear equations. Write each answer as the key formula or value in symbols using the on-screen math keyboard (use the Algebra tab for exponents, fractions, and radicals).

1. Write the distributive property for a(b + c).
2. Evaluate 3 + 4 · 2^2.
3. Solve 3x − 7 = 11.
4. Solve 5x + 3 = 2x − 9.

Homework 1.2 — Formulas and proportions

Use the math keyboard for fractions, exponents, and radicals.

1. Solve d = rt for t.
2. Solve F = (9/5)C + 32 for C.
3. Write the cross-product rule for a/b = c/d.
4. Solve 3/7.50 = 8/x.

Midterm — Weeks 1 & 2

Cumulative midterm on expressions, equations, inequalities, functions, and linear graphs. 60 minutes. Math keyboard available; pasting disabled.

1. Solve −3x > 12.
2. Solve |2x − 1| = 7.
3. Rewrite |A| < k as a compound inequality.
4. Evaluate f(4) for f(x) = 2x^2 − 3.
5. Write the slope formula through (x_1, y_1) and (x_2, y_2).
6. Write the slope-intercept form of a line.
7. Write the point-slope form of a line through (x_1, y_1).
8. Write the condition on slopes m_1 and m_2 for perpendicular lines.

Final Exam — Algebra 1

Cumulative final covering all four weeks. 90 minutes. Math keyboard available; pasting disabled.

1. Solve the system 2x + 3y = 12, 4x − 3y = 6.
2. Write the product rule for exponents, a^m · a^n.
3. Write the definition of a^(−n).
4. Write the general exponential function with initial value a.
5. Expand (a + b)^2.
6. Factor a^2 − b^2.
7. Write the zero-product property.
8. Write the axis of symmetry of y = ax^2 + bx + c.
9. Write the quadratic formula.
10. Write the Pythagorean theorem for legs a, b and hypotenuse c.

Answer key

Homework 1.1 — Expressions and equations:

1. a(b + c) = ab + ac
2. 19
3. x = 6
4. x = −4

Homework 1.2 — Formulas and proportions:

1. t = d / r
2. C = (5/9)(F − 32)
3. ad = bc
4. x = 20

Midterm — Weeks 1 & 2:

1. x < −4
2. x = 4 or x = −3
3. −k < A < k
4. f(4) = 29
5. m = (y_2 − y_1) / (x_2 − x_1)
6. y = mx + b
7. y − y_1 = m(x − x_1)
8. m_1 · m_2 = −1

Final Exam — Algebra 1:

1. x = 3, y = 2
2. a^m · a^n = a^(m+n)
3. a^(−n) = 1 / a^n
4. y = a · b^x
5. (a + b)^2 = a^2 + 2ab + b^2
6. a^2 − b^2 = (a + b)(a − b)
7. ab = 0 ⇒ a = 0 or b = 0
8. x = −b / (2a)
9. x = [−b ± √(b^2 − 4ac)] / (2a)
10. a^2 + b^2 = c^2